

Hussain-Sehgal Theorem - Most General Framework

HUSSAIN-SEHGAL THEOREM (1975)

The most general form of contractions encompassing Ćirić and others.

STATEMENT:

Let (X, d) be a complete metric space and $T: X \rightarrow X$ a mapping of X into itself. Let $\varphi: (\mathbb{R}^+)^5 \rightarrow \mathbb{R}^+$ be a continuous function that is nondecreasing in each coordinate variable, and let T satisfy:

$$d(Tx, Ty) \leq \varphi(d(x, Tx), d(y, Ty), d(x, Ty), d(y, Tx), d(x, y))$$

for all $x, y \in X$.

If $\varphi(t, t, a_1t, a_2t, a_5t) < t$ for all $t > 0$, where $a_1 \in \{0, 1, 2\}$, $a_2 \in \{0, 1, 2\}$, and $a_1 + a_2 \leq 2$, then T has a unique fixed point $u \in X$.

Moreover, for any $x \in X$, the Picard iteration sequence $\{T^n x\}$ converges to u .

EXAMPLES OF VALID φ :

- $\varphi(t_1, t_2, t_3, t_4, t_5) = a_1t_1 + a_2t_2 + a_3t_3 + a_4t_4 + a_5t_5$ (Hardy-Rogers)
- $\varphi(t_1, t_2, t_3, t_4, t_5) = \max\{t_5, (t_1 + t_2)/2, (t_3 + t_4)/2\}$ (Zamfirescu)
- $\varphi(t_1, t_2, t_3, t_4, t_5) = \sqrt{(t_1 \cdot t_2)} + t_5/4$ (Mixed)

ADVANTAGES:

- ✓ Covers all classical contractions as special cases
- ✓ Allows for discontinuous mappings
- ✓ Provides convergence guarantees
- ✓ Flexible enough for practical applications